

Welfare & Efficiency: Taxation

4. For the market of coffee the demand function is $P=120-3Q$ and supply function is $P=30+2Q$. If the government imposes a \$15 tax on the sellers of coffee, calculate the following:

- a) New equilibrium price buyers pay, price sellers receive, and quantity sold.
- b) New Consumer Surplus (CS) and Producer Surplus (PS) after tax.
- c) Tax Revenue, Deadweight Loss (DWL).
- d) Draw a properly labelled diagram based on your answers.

Initially Find the equilibrium price & quantity

$$120 - 3Q = 30 + 2Q$$

$$\therefore Q = 18 \quad \therefore P = \$66$$

Maximum price, $Q=0$ at D	Minimum price, $Q=0$ at S
$P = 120 - 3(0) \quad \therefore P = \120	$P = 30 + 2(0) \quad \therefore P = \30

Before-tax

$$CS = \frac{1}{2} \times B \times H = \frac{1}{2} \times 18 \times (120 - 66) = \$486$$

$$PS = \frac{1}{2} \times B \times H = \frac{1}{2} \times 18 \times (66 - 30) = \$324$$

$$TS = CS + PS = \$810$$

After-tax

$$\text{new S: } P = 30 + 2Q + 15 \quad \therefore \text{new S: } P = 45 + 2Q$$

2) To Find new equilibrium quantity

$$\text{new S} = \text{old D}$$

$$120 - 3Q = 45 + 2Q$$

$$\therefore Q = 15 \rightarrow \text{new } Q$$

• To find price buyers pay (P_B)

Insert new Q in D

$$P_B = 120 - 3(15) \quad \therefore P_B = \$75$$

• To find price sellers receive (P_S)

Insert new Q in old S

$$P_S = 30 + 2(15) \quad \therefore P_S = \$60$$

b) new CS = $\frac{1}{2} \times B \times H = \frac{1}{2} \times 15 \times (120 - 75) = \337.5

new PS = $\frac{1}{2} \times B \times H = \frac{1}{2} \times 15 \times (60 - 30) = \225

Annotations: new Q, Max price, P_D , P_S , Min Price

c) Tax Revenue = Tax $\times$ new Q = $15 \times 15 = \$225$

DWL = $\frac{1}{2} \times B \times H = \frac{1}{2} \times (75 - 60) \times (18 - 15) = \3.5

Annotations: Tax, quantity lost

